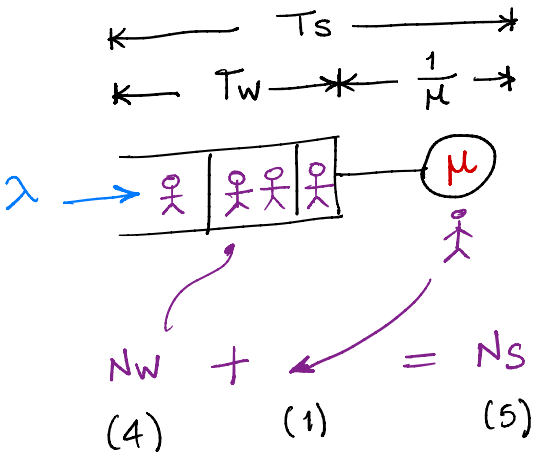
Herramientas Matemáticas

Sesión 10

22 de octubre de 2026



Medidas de desempeño:

↳ $E[\cdot]$

Para el sistema M/M/1:

$$P[N_s = n] = P_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right)$$

$E[N_s]$ → Número promedio de usuarios en el sistema o tamaño promedio de la cola.

$$E[N_s] = \sum_{n=0}^{\infty} n\, P[N_s = n]$$

$$E[N_s] = \sum_{n=0}^{\infty} n\, A^n (1 - A), \text{ con } A = \frac{\lambda}{\mu}$$

$$= (1 - A) \sum_{n=0}^{\infty} n A^n$$

$n x^n$

$$\frac{d}{dx} x^n = n x^{n-1}$$

$$E[N_s] = (1 - A) A \sum_{n=0}^{\infty} n A^{n-1}$$

$$E[N_s] = (1 - A) A \sum_{n=0}^{\infty} \frac{d}{dA} A^n$$

$$E[N_s] = (1 - A) A \frac{d}{dA} \underbrace{\sum_{n=0}^{\infty} A^n}_{S_\infty}$$

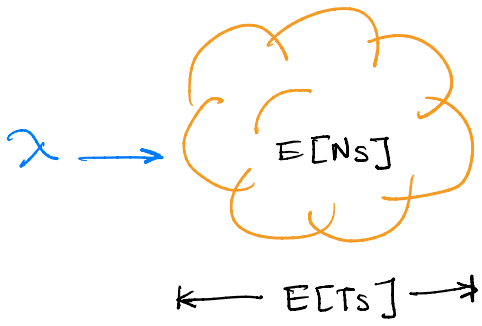
$$E[N_s] = (1 - A) A \frac{d}{dA} \frac{1}{1 - A}$$

$$E[N_s] = \frac{(1 - A) A (-1)(-1)}{(1 - A)^2}$$

Como $\lambda < \mu$: $A < 1$

$$E[N_s] = \frac{A}{1 - A} = \frac{\frac{\lambda}{\mu}}{1 - \frac{\lambda}{\mu}} = \frac{\lambda}{\mu - \lambda}$$

Teorema de Little:



$$\lambda = \frac{E[Ns]}{E[Ts]}$$

$E[Ts] = 2$ horas

$E[Ns] = 68$ km

$$\lambda = \frac{68\ km}{2\ h} = 34\ km/h$$

Para la cola M/M/1

$$E[Ns] = \frac{\lambda}{\mu - \lambda}$$

$$\lambda = \frac{\frac{\lambda}{\mu-\lambda}}{E[Ts]}$$

$$E[Ts] = \frac{1}{\mu - \lambda}$$

Como $E[Ts] = E[Tw] + \frac{1}{\mu}$

$$\frac{1}{\mu-\lambda} = E[Tw] + \frac{1}{\mu}$$

$$E[Tw] = \frac{1}{\mu-\lambda} - \frac{1}{\mu}$$

$$E[Tw] = \frac{\mu - \mu + \lambda}{(\mu-\lambda)\mu}$$

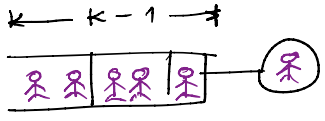
$$E[Tw] = \frac{\lambda}{\mu(\mu-\lambda)}$$

"Tiempo promedio de espera"

Medidas de desempeño para los sistemas:

1) M/M/1/K

$$P_n = \frac{(1-A)A^n}{1-A^{K+1}}$$



$$P_K = \frac{(1-A)A^K}{1-A^{K+1}}$$ ← Probabilidad de bloqueo.

2) M/M/K/K

$$P_n = \frac{\frac{A^n}{n!}}{\sum_{n=0}^{K} \frac{A^n}{n!}}$$

$$P_{K+1} = \frac{\frac{A^{K+1}}{(K+1)!}}{\sum_{n=0}^{K+1} \frac{A^n}{n!}}$$

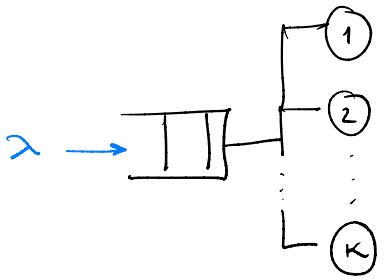
$$P_K = \frac{\frac{A^K}{K!}}{\sum_{n=0}^{K} \frac{A^n}{n!}} \leftarrow \text{probabilidad de bloqueo}$$

$$P_{K+1} = \frac{\frac{A^K A}{(K+1)K!}}{\sum_{n=0}^{K} \frac{A^n}{n!} + \frac{A^{K+1}}{(K+1)!}}$$

Con $K = 1$

$$P_1 = \frac{\frac{A^1}{1!}}{\sum_{n=0}^{1} \frac{A^n}{n!}} = \frac{A}{\frac{A^0}{0!} + \frac{A^1}{1!}} = \frac{A}{1+A}$$

3) M/M/K/∞



1) $E_{1,K} \leftarrow P_K$

2) $E_{2,K} = \frac{K P_K}{K + A(P_K - 1)}$

3) $E[T_W] = \frac{E_{2,K}}{\mu(K-A)}$

Relación entre:

$E_{2,K}$ ↙

$P[T_W > 0]$ de Erlang C

y P_K de Erlang B.

↖ $E_{1,K}$

Actividad 2 del trabajo grupal:

tea = [	1	2	3	7	2	3	8	0	5]
tds = [	9	1	4	1	1	5	3	8	2]

reloj = [	1	3	6	13	15	18	26	26	31]
inicio = [	1	10	11	15	16	18	26	29	37]
salida = [	10	11	15	16	17	23	29	37	39]
espera = [	0	7	5	2	1	0	0	3	6]

